Data Loading (ETL) Assignment

Q1. Data Understanding

The dataset contains several data quality issues that could cause problems during data loading:

- `Order_ID` O101 appears more than once, creating a duplicate.
- O102 has a missing `Sales_Amount`, shown as `Null`.
- O104 has `Three Thousand` in `Sales_Amount`, which is text rather than a number.
- The `Order_Date` values use different formats: `DD-MM-YYYY` and `YYYY/MM/DD`.
- Dates may be interpreted incorrectly if the expected date format is not specified.

Q2. Primary Key Validation

Assuming `Order_ID` is the primary key:

a) Is the dataset violating the primary key rule?
Yes. A primary key must contain a unique value for each record.

b) Which record causes this violation?
O101 appears twice. Both records have Customer_ID C001, Sales_Amount 4500, and Order_Date 12-01-2024.

Q3. Missing Value Analysis

The `Sales_Amount` column contains a missing value.

a) Affected record:
O102 has a missing `Sales_Amount`, shown as `Null`.

b) Why is loading this record without handling the missing value risky?
The database may reject the record if the column does not allow null values. If the record is loaded with a null amount, it may be excluded from sales totals. If the missing amount is replaced with zero without verification, the reports may show incorrect sales.

Q4. Data Type Validation

a) Which record will fail numeric validation?
O104 will fail because its `Sales_Amount` is `Three Thousand`, which is text and not a numeric value.

b) What would happen if the dataset were loaded into a SQL table where `Sales_Amount` is `DECIMAL`?
The database would be unable to convert `Three Thousand` to a decimal number. It may reject the record or stop the load, depending on the loading method and error-handling settings.

Q5. Date Format Consistency

a) Date formats present in the dataset:

- `DD-MM-YYYY`, for example `12-01-2024` and `15-01-2024`
- `YYYY/MM/DD`, for example `2024/01/18`

b) Why is this a problem during data loading?
The loading process may parse dates in one format but reject dates in another. Dates can also be interpreted incorrectly if the format or locale is ambiguous. This could place records in the wrong reporting period.

Q6. Load Readiness Decision

a) Should this dataset be loaded directly into the database?
No.

b) Justification:

1. O101 is duplicated, violating the assumed primary key rule.
2. O102 has a missing sales amount.
3. O104 has a text value in a column that should contain numbers.
4. The dates use inconsistent formats and should be standardized before loading.

Q7. Pre-Load Validation Checklist

Before loading the dataset, I would perform these checks:

1. Confirm that the column names and data types match the target table.
2. Check that every `Order_ID` is present and unique.
3. Check required columns for missing or blank values.
4. Confirm that every `Sales_Amount` is numeric and within the allowed range.
5. Identify text placeholders such as `Null` and convert them to actual null values.

6. Check that every `Order_Date` can be parsed using the agreed date format.
7. Check that dates fall within the expected range.
8. Identify duplicate records and confirm how they should be handled.
9. Compare the source row count with the counts of valid, rejected, and duplicate records.
10. Record any validation errors so they can be corrected and reviewed.

Q8. Cleaning Strategy

1. Keep a copy of the original dataset for reference.
2. Convert text placeholders such as `Null` into actual missing values.
3. Investigate the missing `Sales_Amount` for O102. Fill it only if the correct value can be verified; otherwise, keep the record aside for review.
4. Investigate the duplicate O101 records. Confirm whether one is accidental, then remove or correct the duplicate according to the source data.
5. Verify whether `Three Thousand` in O104 means 3000. If verified, convert it to a numeric value; otherwise, set the record aside for review.
6. Convert all dates to one standard date format, such as `YYYY-MM-DD`.
7. Run the validation checks again after cleaning.
8. Load the validated records and keep a record of any rejected rows and their reasons.

Q9. Loading Strategy Selection

a) Should a Full Load or Incremental Load be used?
An incremental load should be used for daily sales data, assuming previous sales data is already in the database.

b) Justification:
An incremental load adds new or changed records without reloading the entire sales history each day. It is more efficient for regular daily updates. The process should include rules to detect duplicates and handle corrections. A one-time full load may be used for the initial loading of historical data.

Q10. BI Impact Scenario

a) What incorrect results might appear in the Total Sales KPI?
The Total Sales KPI could be overstated if O101 is counted twice. It could be understated if O102's missing amount is excluded or treated as zero. The refresh may also fail or omit O104 if its amount cannot be converted to a number. Inconsistent dates could cause sales to appear in the wrong reporting period.

b) Which records could cause misleading insights?

- O101 could inflate sales because it appears twice.
- O102 could reduce or distort totals because its sales amount is missing.
- O104 could be rejected or excluded because its sales amount is text.
- O103 could be placed in the wrong period if its date is parsed incorrectly.

c) Why might BI tools not detect these issues automatically?
BI tools generally use the data they receive and may not know the business rules for the dataset. They may count duplicate rows as separate transactions, omit missing values from totals, or rely on the loading process to convert dates and numbers. The rules for unique order IDs and complete sales amounts must be defined and checked as part of the data preparation process.